

Ye Botao

AI Researcher & Builder



About

Currently focused on medical text hallucination detection — building systems that make AI trustworthy in healthcare. I believe in the compounding effect: invest time where growth becomes exponential. True resilience isn't dependence on rules — it's understanding their logic. Aspiring entrepreneur since high school. I value the freedom to create above all else.

Opportunities exist where the concepts have not yet been invented.

— YE BOTAO

Projects

ONGOING RESEARCH

2025

CuraView - Medical AI Hallucination Detection & Correction

Multi-agent hallucination detection for medical LLMs. GraphRAG + Qwen + MS-Swift, 46,998 MIMIC-IV records. Evidence-driven error synthesis and correction.

Hallucination Detection

Knowledge Graph Reasoning

RL Fine-tuning Pipeline

Medical Synthetic Data

On-device Model Training

DIGITAL HUMANITIES

2025

Narrative-Causal-Graph - Causal Event Network for Stories

Novels into structured causal event graphs. 4,700 events, 3,200 causal links. Neo4j + CSV + JSON-LD output for quantitative literary analysis.

Narrative Causal Mining

Literary Theory-Driven

Structured Knowledge Extraction

Graph DB Generation

BUSINESS ANALYTICS

2026

OpenStock Analyst - Intelligent Stock Analysis Platform

Four-layer ranking framework (valuation/profitability/safety/growth) generates structured reports for US stocks, HK stocks, and crypto. F-Score breakdown and scenario analysis.

Quantitative Value Investing

Rank-Driven Decisions

Anti-Hallucination Engineering

Cross-Asset Comparison

CUSTOMER ANALYTICS

2026

Decathlon VOC Analyzer - Voice of Customer Intelligence

Product text, images, and reviews into evidence-backed VOC reports. Claim-level citations, auditable multimodal customer insights.

Multimodal Customer Insights

Evidence Traceability

Audit Trail

Graceful Degradation

SECURITY RESEARCH

2026

Security-Tax - Multi-Agent Security Defense System

LangChain multi-agent security defense. 7 agents, dual-layer defense, 12 jailbreak attacks. Interactive flow visualization and batch comparison.

Agent Security Attack/Defense

Safety-Collaboration Tradeoff

Propagation Attack Simulation

Comparative Experiment Framework

ENERGY AI
2026

Energy Demand Prediction
- Causal Explainable AI

CNN-LSTM-Attention parallel architecture, 34.84% improvement over serial. Bayesian network causal explanations (0.999+ cosine similarity). Visualization and energy-saving recommendations.

- Deep Causal Reasoning
- Feature-Causal Bridge
- Stable Explainability
- Counterfactual Analysis

AI APPLICATION
2025

NutriFlow AI - Intelligent
Nutrition Analysis

LangChain + Qwen multimodal nutrition analysis. Meal image recognition, health scoring, 12 collaborative tools, personalized recommendations.

- End-to-End Diet Perception
- Vision-Knowledge Synergy
- Autonomous Agent Pipeline
- Multi-Dim Health Assessment

CONTRACT WORK
2023

AI Review Assistant -
Medical Document
Processing

First professional contract. Windows desktop medical RAG proofreading tool, GPT-4o + Qwen, batch document error detection.

- Medical Document Proofreading
- Structured AI Output
- Human-AI Interactive Review
- Desktop Deployment

STARTUP COLLAB
2025

GReSy - Game Player
Matching System

Startup project for Imperial College graduates. Multi-dimensional matching for game preferences + MBTI, five strategy patterns. Equity offer received.

- Multi-Dim Preference Matching
- Strategy Pattern Decoupling
- Game-Theoretic Scoring
- Geo-Aware Grouping

COMPETITION
EY ESG Challenge

Team Leadership - Notion
Workflow System

EY ESG competition team lead. Notion full-cycle project management system. Praised for logical clarity and rigor.

- Full-Cycle Project Management
- ESG Competition

Stories

A Chance That Would Have Passed Me By

Met a medical journal editor on a train. Discussed AI in publishing. Completed needs analysis, feasibility study, and framework design. Landed my first engineering contract.

Opportunities don't always appear as "projects." They start as a conversation, a pain point, or an undefined problem. What matters is recognizing it and breaking it down into something deliverable.

Catch It, Learn It, Deliver It

Met two Imperial College grads at badminton. They needed a recommendation system. I said: "Give me two weeks." Demo delivered in two weeks. Equity offer in one month.

Learning's value isn't mastering concepts — it's transforming new knowledge into something others actually need and can use.

No Reply? Knock on the Door

Emailed a professor multiple times. No reply. Asked a friend for the office address. Showed up in person. Got the lab offer after a real conversation.

Research and entrepreneurship share this: opportunities require active pursuit. They don't appear automatically — you put yourself in front of the problem and the people first, then prove yourself through clear expression and real action.

Skills

AI & MACHINE LEARNING

LLMs GraphRAG LangChain Multi-Agent LoRA/QLoRA NLP GRPO RAG Prompt Eng Fine-tuning XAI
Bayesian Net Security AI

DEVELOPMENT

Python JavaScript Web Dev Microservices Desktop Apps CI/CD

TOOLS & FRAMEWORKS

PyTorch TensorFlow HuggingFace MS-Swift Ollama LanceDB Notion Git Linux Docker

LANGUAGES

Chinese (Native) English (Fluent) Korean

SOFT SKILLS

Leadership Team Building Project Management Fast Learning Initiative

Contact

Email 6severin9@gmail.com

GitHub github.com/severin-ye

I don't aim to be the smartest person in the room —

I aim to be the one who dares to act.

To keep learning, keep moving forward, and keep creating in a complex and ever-changing world.